Epic-FHIR — API category: CDS Hooks (sections 7 – 7.2.6)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

7 API category: CDS Hooks

File: site/apis/07-CDS-Hooks/index.html

R4 (9)

API | Method | URL template | Type
CDS Hooks Condition.Create (Encounter Diagnosis) (R4) | | | Industry-Standard
CDS Hooks Condition.Create (Problems) (R4) | | | Industry-Standard
CDS Hooks MedicationRequest.Create (Unsigned Order) (R4) | | | Industry-Standard
CDS Hooks MedicationRequest.Delete (Unsigned Order) (R4) | | | Industry-Standard
CDS Hooks MedicationRequest.Read (Unsigned Order) (R4) | GET | /api/FHIR/R4/MedicationRequest/{ID} | Industry-Standard
CDS Hooks ServiceRequest.Create (Unsigned Order) (R4) | | | Industry-Standard
CDS Hooks ServiceRequest.Delete (Unsigned Order) (R4) | | | Industry-Standard
CDS Hooks ServiceRequest.Read (Unsigned Order) (R4) | GET | /api/FHIR/R4/ServiceRequest/{ID} | Industry-Standard
CDS Hooks ServiceRequest.Update (Unsigned Order) (R4) | | | Industry-Standard

STU3 (6)

API | Method | URL template | Type
CDS Hooks Condition.Create (Encounter Diagnosis, Problems) (STU3) | | | Industry-Standard
CDS Hooks MedicationRequest.Create (Unsigned Order) (STU3) | | | Industry-Standard
CDS Hooks MedicationRequest.Delete (Unsigned Order) (STU3) | | | Industry-Standard
CDS Hooks ProcedureRequest.Create (Unsigned Order) (STU3) | | | Industry-Standard
CDS Hooks ProcedureRequest.Delete (Unsigned Order) (STU3) | | | Industry-Standard
CDS Hooks ProcedureRequest.Update (Unsigned Order) (STU3) | | | Industry-Standard

7.1 CDS Hooks: R4

File: site/apis/07-CDS-Hooks/1-R4/index.html

* CDS Hooks Condition.Create (Encounter Diagnosis) (R4)
* CDS Hooks Condition.Create (Problems) (R4)
* CDS Hooks MedicationRequest.Create (Unsigned Order) (R4)
* CDS Hooks MedicationRequest.Delete (Unsigned Order) (R4)
* CDS Hooks MedicationRequest.Read (Unsigned Order) (R4)
* CDS Hooks ServiceRequest.Create (Unsigned Order) (R4)
* CDS Hooks ServiceRequest.Delete (Unsigned Order) (R4)
* CDS Hooks ServiceRequest.Read (Unsigned Order) (R4)
* CDS Hooks ServiceRequest.Update (Unsigned Order) (R4)

7.1.1 CDS Hooks Condition.Create (Encounter Diagnosis) (R4)

File: site/apis/07-CDS-Hooks/1-R4/1058-CDS-Hooks-Condition-Create-Encounter-Diagnosis-R4.html
Epic API ID | 1058
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1058

Description

This web service adds encounter diagnoses to a patient's chart. This web service is currently designed only for use with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Remarks

{

```
  "problems":"Condition?patient={{context.patientId}}&clinical-status=active&category=problem-list-item",
  "encounterDx":"Condition?patient={{context.patientId}}&category=encounter-diagnosis&encounter={{context.encounterId}}"
}
```

Sample request

```
{
  "hookInstance": "e5abb534-ebdd-11eb-9b60-0050568b7be6",
  "fhirServer": "https://hostname/instance/api/FHIR/R4",
  "hook": "patient-view",
  "fhirAuthorization": {
    "access_token": "<opaque access token>",
    "token_type": "Bearer",
    "expires_in": 3600,
    "scope": "user/Condition.Read user/Condition.Write user/Patient.Read user/Practitioner.Read
    user/PractitionerRole.Read",
    "subject": "<your client_id>"
  },
  "context": {
    "patientId": "eZ5-7rYdWqgv3jSgIvx.SPw3",
    "encounterId": "eUNhr0pVHo.9Lau9LJJbgHQ3",
    "userId": "Practitioner/evNp-KhYwOOqAZn1pZ2enuA3"
  },
  "prefetch": {
    "problems": {
      "resourceType": "Bundle",
      "type": "searchset",
      "total": 2,
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Condition?patient=eZ5-7rYdWqgv3jSgIvx.SPw3&clinical-
    status=active&category=problem-list-item"
        }
      ],
      "entry": [
        {
          "link": [
            {
              "relation": "self",
              "url": "https://hostname/instance/api/FHIR/R4/Condition/eBUgclBp0IO8k6haU7cfXGg3"
            }
          ],
          "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eBUgclBp0IO8k6haU7cfXGg3",
          "resource": {
            "resourceType": "Condition",
            "id": "eBUgclBp0IO8k6haU7cfXGg3",
            "clinicalStatus": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
                  "version": "4.0.0",
                  "code": "active",
                  "display": "Active"
                }
              ],
              "text": "Active"
            },
            "verificationStatus": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
                  "version": "4.0.0",
                  "code": "confirmed",
                  "display": "Confirmed"
                }
              ],
              "text": "Confirmed"
            },
            "category": [
              {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/condition-category",
                    "code": "problem-list-item",
                    "display": "Problem List Item"
                  }
                ],
                "text": "Problem List Item"
              }
            ],
            "code": {
              "coding": [
                {
                  "system": "urn:oid:2.16.840.1.113883.6.90",
                  "code": "E03.9"
                },
                {
                  "system": "urn:oid:2.16.840.1.113883.6.96",
```

```
          "code": "40930008"
        }
      ],
      "text": "Hypothyroidism"
    },
    "subject": {
      "reference": "Patient/eZ5-7rYdWqgv3jSgIvx.SPw3",
      "display": "Ambulatory, Oliver"
    },
    "onsetPeriod": {
      "start": "2008-04-17",
      "end": "2008-04-17"
    },
    "recordedDate": "2008-04-17"
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/e0lOm4N9POJvVqbipHCSBrw3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/e0lOm4N9POJvVqbipHCSBrw3",
  "resource": {
    "resourceType": "Condition",
    "id": "e0lOm4N9POJvVqbipHCSBrw3",
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
          "version": "4.0.0",
          "code": "active",
          "display": "Active"
        }
      ],
      "text": "Active"
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "problem-list-item",
            "display": "Problem List Item"
          }
        ],
        "text": "Problem List Item"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.6.90",
          "code": "E78.00"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.6.96",
          "code": "13644009"
        }
      ],
      "text": "Hypercholesterolemia"
    },
    "subject": {
      "reference": "Patient/eZ5-7rYdWqgv3jSgIvx.SPw3",
      "display": "Ambulatory, Oliver"
    },
    "onsetPeriod": {
      "start": "2010-12-23",
      "end": "2010-12-23"
    },
    "recordedDate": "2010-12-23"
  },
  "search": {
    "mode": "match"
```

```
        }
      }
    ]
  },
  "encounterDx": {
    "resourceType": "Bundle",
    "type": "searchset",
    "total": 0,
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Condition?patient=eZ5-7rYdWqgv3jSgIvx.SPw3&category=encounter-
diagnosis&encounter=eUNhr0pVHo.9Lau9LJJbgHQ3"
      }
    ],
    "entry": [
      {
        "resource": {
          "resourceType": "OperationOutcome",
          "issue": [
            {
              "severity": "warning",
              "code": "processing",
              "details": {
                "coding": [
                  {
                    "system": "urn:oid:1.2.840.114350.1.13.0.1.7.2.657369",
                    "code": "4101",
                    "display": "Resource request returns no results."
                  }
                ],
                "text": "Resource request returns no results."
              }
            }
          ]
        },
        "search": {
          "mode": "outcome"
        }
      }
    ]
  }
 },
 "extension": {
   "com.epic.cdshooks.request.bpa-trigger-action": "92",
   "com.epic.cdshooks.request.cds-hooks-specification-version": "1.0",
   "com.epic.cdshooks.request.fhir-version": "R4",
   "com.epic.cdshooks.request.criteria-id": "335",
   "com.epic.cdshooks.request.epic-version": "9.7",
   "com.epic.cdshooks.request.cds-hooks-implementation-version": "1.1",
   "com.epic.cdshooks.request.cds-hooks-su-version": "0",
   "internal.epic.cdshooks.request.epic-emp-id": "10000"
 }
}
```


Sample response


```
{
    "cards": [{
            "detail": "Other primary thrombophilia not documented as an encounter diagnosis. Consider adding it to the
    chart (and optionally the problem list).",
            "indicator": "warning",
            "links": [{
                    "appContext": "e5abb534-ebdd-11eb-9b60-0050568b7be6 (or any other string)",
                    "label": "Test Smart App Link",
                    "type": "smart",
                    "url": "https://test.smartapp.com/launch"
                }
            ],
            "selectionBehavior": "any",
            "source": {
                "label": "Other primary thrombophilia screening service"
            },
            "suggestions": [{
                    "label": "Add Other primary thrombophilia as encounter Dx [D68.59]",
                    "actions": [{
                            "description": "Add Other primary thrombophilia as encounter Dx [D68.59]",
                            "resource": {
                                "category": [{
                                        "coding": [{
                                                "code": "encounter-diagnosis",
                                                "system": "http://terminology.hl7.org/CodeSystem/condition-category"
                                            }
                                        ],
                                        "text": "Encounter Diagnosis"
                                    }
                                ],
                                "code": {
                                    "coding": [{
```

```
                                        "code": "D68.59",
                                        "system": "urn:oid:2.16.840.1.113883.6.90"
                                    }
                                ],
                                "text": "Other primary thrombophilia" //Either text or code.coding.display is
    required.
                            },
                            "subject": {
                                "reference": "Patient/eZ5-7rYdWqgv3jSgIvx.SPw3"
                            },
                            "resourceType": "Condition",
                            "encounter": {
                                "reference": "Encounter/eUNhr0pVHo.9Lau9LJJbgHQ3"
                            }
                        },
                        "type": "create"
                    }
                ],
                "isRecommended": true,
                "uuid": "<your uuid>"
            }
        ]
    }
]
}
```

7.1.2 CDS Hooks Condition.Create (Problems) (R4)

File: site/apis/07-CDS-Hooks/1-R4/1057-CDS-Hooks-Condition-Create-Problems-R4.html
Epic API ID | 1057
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
Keywords | Problems;Problem List;Condition;Diangosis
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1057

Description

This web service adds problems to a patient's chart. This web service is currently designed only for use with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

This web service works similar to the Condition.Create (Problems) (R4) web service except it does not currently support the following fields: onsetDateTime, onSetPeriod, clinicalStatus, text.

Sample response

```
{
    "cards": [{
            "summary": "CDS Hooks Card",
            "detail": "Details for CDS Hooks Card",
            "uuid": "UUID for Card",
            "source": {
                "label": "Sourced from Epic's Vendor Services Unit Test endpoint",
                "icon": "https://www.epic.com/favicon.ico",
                "url": "https://appmarket.epic.com/api/TestCDSHooks"
            },
            "links": [{
                    "label": "SMART App",
                    "url": "https://appmarket.epic.com/Test/smart",
                    "type": "smart",
                    "appContext": "b516b58a-dbde-4888-a8a2-24173d63adcb" //Context to tie CDS instance to SMART on FHIR
    app launch
                }
            ],
            "suggestions": [{
                    "label": "Create Problem List Condition", //required
                    "uuid": "create-condition",  //Should be implemented as guid
                    "isRecommended": false,  //Determines if automatically selected as follow-up or if user has to
    manually select problem list suggestion
                    "actions": [{
                            "type": "create",  //always "create"
                            "description": "Create Problem List Dx",  //Used as display next to "Add Problem"/"Do Not
    Add" next to description of diagnosis obtained from code mapping
                            "resource": {
                                "code": {
                                    "text": "Other primary thrombophilia",  //unused
                                    "coding": [{
```

```
                                        "code": "D68.59",  //coding is mapped to diagnosis data in Epic
                                        "system": "urn:oid:2.16.840.1.113883.6.90"
                                    }
                                ]
                            },
                            "category": [{  //Required element
                                    "text": "Problem List Item",
                                    "coding": [{
                                            "code": "problem-list-item",
                                            "system": "http://terminology.hl7.org/CodeSystem/condition-category"
                                        }
                                    ]
                                }
                            ],
                            "resourceType": "Condition",  //Required
                            "subject": {
                                "reference": "Patient/el0VeGTE1vcmdn5N.BKPUFg3"  //Patient FHIR ID required
                            }
                        }
                    }
                ]
            }
        ]
    }
]
}
```

7.1.3 CDS Hooks MedicationRequest.Create (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/1060-CDS-Hooks-MedicationRequest-Create-Unsigned-Order-R4.html
Epic API ID | 1060
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1060

Description

This web service adds unsigned medication orders for a patient. This web service is currently designed only for use with CDS Hooks integrations and should not be used for other purposes. When used in a CDS Hooks context, this web service suggest follow-up medications orders in the CDS card displayed to the user. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Sample response

```
{
  "cards": [
    {
      "summary": "CDS Hooks example card summary",
      "selectionBehavior": "at-most-one",
      "suggestions": [
        {
          "label": "Add these actions",
          "actions": [
            {
              "type": "create",
              "description": "Test Medication Order",
              "resource": {
                "resourceType": "MedicationRequest",
                "status": "draft",
                "intent": "proposal",
    "category": [
                  {
                    "coding": [
                      {
                        "system": "http://terminology.hl7.org/CodeSystem/medicationrequest-category",
                        "code": "inpatient",
                        "display": "Inpatient"
                      }
                    ]
                  }
                ],
                "medicationCodeableConcept": {
                  "coding": [
                    {
                      "system": "urn:com.epic.cdshooks.action.code.system.preference-list-item",
                      "code": "IBU"
                    }
```

```
                ],
                "text": "Ibuprofen 400 MG PO TABS"
              },
    "reasonReference": [
      {
        "reference": "Condition/ejwkKH6vIAe5l8Tcdlz1HJbBhhHptXy0VPfxVihFyfMIYuZyi5MBLePwAI.QLCSVuQMgm-
    SZww1QNfUs.S5t5xQ3",
        "type": "Condition",
        "display": "Cough"
      },
      {
        "reference": "Condition/ejwkKH6vIAe5l8Tcdlz1HJbBhhHptXy0VPfxVihFyfMJTnDUPRmH6rFLa9S0PUX00KNXN-
    YhAmGzPS4Pq.3u-ig3",
        "type": "Condition",
        "display": "Stomach pain"
      }
    ],
    "contained": [
      {
      "resourceType": "Observation",
      "id": "Indication1",
      "status": "preliminary",
      "code": {
        "text": "Indications Of Use"
      },
      "valueString": "headache"
      },
      {
      "resourceType": "Observation",
      "id": "Indication2",
      "status": "preliminary",
      "code": {
        "text": "Indications Of Use"
      },
      "valueString": "sneeze"
      },
      {
      "resourceType": "Observation",
      "id": "IndicationsComment",
      "status": "preliminary",
      "code": {
        "text": "Indications Of Use Comment"
      },
      "valueString": "Comment"
      },
      {
      "resourceType": "QuestionnaireResponse",
      "id": "OrderSpecificQuestions",
      "status": "in-progress",
      "item": [
        {
        "linkId": "Sample Question 1",
        "answer": [
          {
          "extension": [
            {
            "valueString": "Sample Answer 1",
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-question-comment"
            }
          ],
          "valueString": "Yes"
          }
        ]
        },
      ]
      }
    ]
    }
  ]               "subject": {
                "reference": "Patient/eorRFcFtkMn3fTS05W1ZlvA3",
                "display": "Argonaut, Jason"
              }
            }
          }
        ]
      }
    ]
  }
]
}
```


7.1.4 CDS Hooks MedicationRequest.Delete (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/10160-CDS-Hooks-MedicationRequest-Delete-Unsigned-Order-R4.html
Epic API ID | 10160
FHIR version | R4
Categories | CDS Hooks

API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10160

Description

Sample request


```
{
  "hookInstance": "9e4c485e-b80e-11ef-977a-005056be447d",
  "fhirServer": "https://hostname/instance/api/FHIR/R4",
  "hook": "order-select",
  "fhirAuthorization": {
    "access_token": "<opaque access token>",
    "token_type": "Bearer",
    "expires_in": 3600,
    "scope": "<application scopes>",
    "subject": "<your client id>"
  },
  "context": {
    "patientId": "el0VeGTE1vcmdn5N.BKPUFg3",
    "encounterId": "ejHSgHTi-oTHAIeAOzTmnCA3",
    "userId": "PractitionerRole/eg55ZEeKE6a5xlFwDbwiNtA3",
    "selections": [
      "MedicationRequest/ewfTRiRrfssgO6Y0r-
    cxLbrvlme2tzPhWUICQq0v7IhGJFfbD7yvK6DREsXrInaq4Lio1uym.c0mciAN90lAo9oxuTjDAJ5Qs5P30I4OruqpxidgkpUa2PFUWxv4CED.-
    I2bYyDTBgWwzWK24pnlDd1eWDhnzr1Ep6beUQ8BDLufGIrOWnwMfCkpH4fVd54VH3"
    ],
    "draftOrders": {
      "resourceType": "Bundle",
      "type": "collection",
      "entry": [
        {
          "resource": {
            "resourceType": "MedicationRequest",
            "id": "ewfTRiRrfssgO6Y0r-
    cxLbrvlme2tzPhWUICQq0v7IhGJFfbD7yvK6DREsXrInaq4Lio1uym.c0mciAN90lAo9oxuTjDAJ5Qs5P30I4OruqpxidgkpUa2PFUWxv4CED.-
    I2bYyDTBgWwzWK24pnlDd1eWDhnzr1Ep6beUQ8BDLufGIrOWnwMfCkpH4fVd54VH3",
            "status": "draft",
            "intent": "order",
            "category": [
              {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/medicationrequest-category",
                    "code": "inpatient",
                    "display": "Inpatient"
                  }
                ],
                "text": "Inpatient"
              }
            ],
            "medicationReference": {
              "reference": "Medication/epB4yoDKZsUL25R7hwVJBNg3",
              "display": "ACETAMINOPHEN 325 MG PO TABS"
            },
            "subject": {
              "reference": "Patient/el0VeGTE1vcmdn5N.BKPUFg3",
              "display": "Willow, Eric E"
            },
            "encounter": {
              "reference": "Encounter/ejHSgHTi-oTHAIeAOzTmnCA3",
              "identifier": {
                "use": "usual",
                "system": "urn:oid:1.2.840.114350.1.13.0.1.7.3.698084.8",
                "value": "15249"
              },
              "display": "Hospital Encounter"
            },
            "requester": {
              "reference": "Practitioner/eM5CWtq15N0WJeuCet5bJlQ3",
              "type": "Practitioner",
              "display": "Physician Family Medicine, MD"
            },
            "recorder": {
              "reference": "Practitioner/eM5CWtq15N0WJeuCet5bJlQ3",
              "type": "Practitioner",
              "display": "Physician Family Medicine, MD"
            },
            "dosageInstruction": [
              {
```

```
        "timing": {
          "repeat": {
            "boundsPeriod": {
              "start": "2024-12-11T22:23:59Z"
            },
            "frequency": 1,
            "period": 6,
            "periodUnit": "h"
          },
          "code": {
            "text": "Every 6 hours as needed"
          }
        },
        "asNeededBoolean": true,
        "route": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "260548002",
              "display": "Oral (qualifier value)"
            },
            {
              "system": "urn:oid:1.2.840.114350.1.13.0.1.7.4.798268.7025",
              "code": "15",
              "display": "Oral"
            }
          ],
          "text": "Oral"
        },
        "doseAndRate": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://epic.com/CodeSystem/dose-rate-type",
                  "code": "calculated",
                  "display": "calculated"
                }
              ],
              "text": "calculated"
            },
            "doseQuantity": {
              "value": 650,
              "unit": "mg",
              "system": "http://unitsofmeasure.org",
              "code": "mg"
            }
          },
          {
            "type": {
              "coding": [
                {
                  "system": "http://epic.com/CodeSystem/dose-rate-type",
                  "code": "admin-amount",
                  "display": "admin-amount"
                }
              ],
              "text": "admin-amount"
            },
            "doseQuantity": {
              "value": 2,
              "unit": "tablet",
              "system": "http://unitsofmeasure.org",
              "code": "{tbl}"
            }
          },
          {
            "type": {
              "coding": [
                {
                  "system": "http://epic.com/CodeSystem/dose-rate-type",
                  "code": "ordered",
                  "display": "ordered"
                }
              ],
              "text": "ordered"
            },
            "doseQuantity": {
              "value": 650,
              "unit": "mg",
              "system": "http://unitsofmeasure.org",
              "code": "mg"
            }
          }
        ]
      }
    ]
  }
},
```

```
{
  "resource": {
    "resourceType": "Medication",
    "id": "epB4yoDKZsUL25R7hwVJBNg3",
    "identifier": [
      {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.0.1.7.2.698288",
        "value": "101"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.6.253",
          "code": "283"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.6.68",
          "code": "64200010000310"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.6.162",
          "code": "03441"
        },
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "161"
        },
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "313782"
        }
      ],
      "text": "acetaminophen (TYLENOL) tablet"
    },
    "form": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.0.1.7.4.698288.310",
          "code": "TABS",
          "display": "tablet"
        }
      ],
      "text": "tablet"
    },
    "ingredient": [
      {
        "itemCodeableConcept": {
          "coding": [
            {
              "system": "urn:oid:2.16.840.1.113883.6.253",
              "code": "283"
            },
            {
              "system": "urn:oid:2.16.840.1.113883.6.68",
              "code": "64200010000310"
            },
            {
              "system": "urn:oid:2.16.840.1.113883.6.162",
              "code": "03441"
            },
            {
              "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
              "code": "161"
            },
            {
              "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
              "code": "313782"
            }
          ],
          "text": "acetaminophen (TYLENOL) tablet"
        },
        "strength": {
          "numerator": {
            "value": 325,
            "unit": "MG",
            "system": "http://unitsofmeasure.org",
            "code": "mg"
          },
          "denominator": {
            "value": 325,
            "unit": "MG",
            "system": "http://unitsofmeasure.org",
            "code": "mg"
          }
        }
      }
    ]
```

```
          }
        }
      ]
    }
  },
  "extension": {
    "com.epic.cdshooks.request.bpa-trigger-action": "18",
    "com.epic.cdshooks.request.cds-hooks-specification-version": "1.1",
    "com.epic.cdshooks.request.fhir-version": "R4",
    "com.epic.cdshooks.request.criteria-id": "504",
    "com.epic.cdshooks.request.epic-version": "11.2",
    "com.epic.cdshooks.request.cds-hooks-implementation-version": "1.8",
    "com.epic.cdshooks.request.cds-hooks-su-version": "0",
    "internal.epic.cdshooks.request.epic-emp-id": "14977"
  }
}
```


Sample response


```
{
  "cards": [
    {
      "uuid": "96918b3c-c068-4f4d-a2af-a573f6194519",
      "summary": "Recommend Removal of Acetaminophen",
      "detail": "This patient is approaching their daily dose of acetaminophen.",
      "indicator": "warning",
      "extension": {
        "com.epic.cdshooks.card.detail.content-type": "text/html"
      },
     "source": {
        "label": "Click here for ACR Radiations Guidelines.",
        "url": "http://www.ncbi.nlm.nih.gov/pubmed/21716271",
    "topic": {
      "code": "AcetaminophenDoseExceeded"
    }
      },
      "suggestions": [
        {
          "uuid": "a2b24553-ae4b-4885-b067-b1e98a2fecc0",
          "label": "Remove Acetaminophen",
          "isRecommended": false,
          "actions": [
            {
              "type": "delete",
              "description": " ",
              <b>"resourceId": "MedicationRequest/ewfTRiRrfssgO6Y0r-
    cxLbrvlme2tzPhWUICQq0v7IhGJFfbD7yvK6DREsXrInaq4Lio1uym.c0mciAN90lAo9oxuTjDAJ5Qs5P30I4OruqpxidgkpUa2PFUWxv4CED.-
    I2bYyDTBgWwzWK24pnlDd1eWDhnzr1Ep6beUQ8BDLufGIrOWnwMfCkpH4fVd54VH3"</b>
            }
          ]
        }
      ],
      "selectionBehavior": "any"
    }
  ]
}
```


7.1.5 CDS Hooks MedicationRequest.Read (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/10031-CDS-Hooks-MedicationRequest-Read-Unsigned-Order-R4.html
Epic API ID | 10031
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/MedicationRequest/{ID}
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10031

Description

This web service reads unsigned medication order information from a patient's chart. This web service is currently designed for use only with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | token | False | | | FHIR identifier for the MedicationRequest.

Response elements

Name | Type | Optional | Array | Cardinality | Description
MedicationRequest | MedicationRequest | False | | | An instance of the R4 MedicationRequest FHIR resource.
MedicationRequest.category | CodeableConcept | False | Yes | | The category of the medication request. Supported categories: inpatient outpatient community discharge
MedicationRequest.contained | ResourceContainer | True | Yes | | Supported resources: Observation QuestionnaireResponse
MedicationRequest.dispenseRequest | BackboneElement | False | | | Parameter group containing information about the dispense.
MedicationRequest.dispenseRequest.expectedSupplyDuration | Duration | False | | | Number of days supply is expected to last.
MedicationRequest.dispenseRequest.numberOfRepeatsAllowed | unsignedInt | True | | | Value for number of refills allowed for this medication.
MedicationRequest.dispenseRequest.performer | Reference | False | | | Reference to the Organization resource.
MedicationRequest.dispenseRequest.quantity | Quantity | False | | | Dispense quantity for the medication.
MedicationRequest.dispenseRequest.validityPeriod | Period | False | | | Dispense duration timeline. Includes start date and end date.
MedicationRequest.dosageInstruction | Dosage | False | Yes | | Dosage instruction details.
MedicationRequest.dosageInstruction.doseAndRate | BackboneElement | False | Yes | | Amount of medication administered, to be administered or typical amount to be administered
MedicationRequest.dosageInstruction.method | CodeableConcept | False | | | Technique for administering medication.
MedicationRequest.dosageInstruction.patientInstruction | string | False | | | Patient or consumer oriented instructions
MedicationRequest.dosageInstruction.route | CodeableConcept | False | | | How drug should enter body
MedicationRequest.dosageInstruction.text | string | False | | | Free text dosage instructions e.g. SIG
MedicationRequest.dosageInstruction.timing | Timing | False | | | Timing information.
MedicationRequest.dosageInstruction.timing.repeat | BackboneElement | False | | | Repeat occurrence information.
MedicationRequest.dosageInstruction.timing.repeat.boundsPeriod | Period | False
MedicationRequest.dosageInstruction.timing.repeat.boundsPeriod.start | String | False | | | Starting time with inclusive boundary
MedicationRequest.dosageInstruction.timing.repeat.count | Int64 | True | | | Number of treatments.
MedicationRequest.dosageInstruction.timing.repeat.countMax | Int64 | True | | | Maximum number of times to repeat
MedicationRequest.dosageInstruction.timing.repeat.dayOfWeek | String | False | | | mon | tue | wed | thu | fri | sat | sun
MedicationRequest.dosageInstruction.timing.repeat.duration | Decimal | True | | | How long when it happens
MedicationRequest.dosageInstruction.timing.repeat.durationUnit | String | False | | | unit of time (UCUM)
MedicationRequest.dosageInstruction.timing.repeat.frequency | Int64 | True | | | Indicates the number of repetitions that should occur within a period.
MedicationRequest.dosageInstruction.timing.repeat.period | decimal | True | | | Interval period.
MedicationRequest.dosageInstruction.timing.repeat.periodUnit | code | False | | | Interval period unit.
MedicationRequest.dosageInstruction.timing.repeat.timeOfDay | String | False | | | Time of day for action
MedicationRequest.dosageInstruction.timing.repeat.when | String | False | | | Code for time period of occurrence
MedicationRequest.encounter | Reference | False | | | Reference to the Encounter resource.
MedicationRequest.identifier | Identifier | False | Yes | | Identifier for the unsigned order.
MedicationRequest.intent | code | False | | | Returns “plan” for historical medications. Returns “order” for other medications.
MedicationRequest.medicationReference | Reference | False | | | Reference to the Medication resource for this unsigned order.
MedicationRequest.extension (also: order-priority-category) | Extension | True | Yes | | The priority of a medication order.
MedicationRequest.priorPrescription | Reference | False | | | Reference to the MedicationRequest resource which the current unsigned order was reordered from.
MedicationRequest.reasonCode | CodeableConcept | True | Yes | | Reason or indication for ordering the medication.
MedicationRequest.reasonReference | Reference | True | Yes | | Reference to the Condition resource.
MedicationRequest.reportedBoolean | boolean | True | | | Determines whether the order is historical.
MedicationRequest.requester | Reference | False | | | Reference to the Practitioner resource.
MedicationRequest.status | code | False | | | Always set to "draft".
MedicationRequest.subject | Reference | False | | | Reference to the Patient resource for the MedicationRequest.
MedicationRequest.supportingInformation | Reference | True | Yes | | Additional clinical information. Reference to contained resources.

Sample request

```
Sample not available
```

Sample response


```
{
  "resource": {
    "resourceType": "MedicationRequest",
    "id":
    "ez067mnwOAKP5z1.YJwRAsdb-2gxXChBAs6WSyFEcPa1cHvcvIwlZ4kRgOp8eHEbXTkaVj0Sgi1L3OKoi.NJvKYs2K-zIeNRqEN8ex2-XbCME-
    vRxCLP41Uiyv5mC1fTKIBPUV2xJEHNqmf0t4Tpm3FoAqt0AkTT8G-n.gZGuGlXzNwYPEiNSuvU5oIlgS2x-3",
   "extension": [
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "urn:oid:1.2.840.114350.1.13.5826.1.7.10.798268.120",
                      "code": "6",
                      "display": "Routine"
                    }
                  ]
                },
                "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-priority-category"
              }
            ],
    "contained": [
```

```
  {
    "resourceType": "Observation",
    "id": "Indication1",
    "status": "preliminary",
    "code": {
      "text": "Indications Of Use"
    },
    "valueString": "MIGRAINE^58"
  },
  {
    "resourceType": "QuestionnaireResponse",
    "id": "OrderSpecificQuestions",
    "status": "in-progress",
    "item": [
      {
        "linkId": "pain",
        "answer": [
          {
            "valueString": "MODERATE"
          }
        ]
      }
    ]
  }
],
"status": "draft",
"intent": "order",
"category": [
  {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/medicationrequest-category",
        "code": "inpatient",
        "display": "Inpatient"
      }
    ],
    "text": "Inpatient"
  }
],
"priority": "routine",
"medicationReference": {
  "reference": "Medication/epu5dX0tg-OkYhXKBX2k3rQ3",
  "display": "IBUPROFEN 400 MG PO TABS"
},
"subject": {
  "reference": "Patient/e6Vyd3Ocr.G6SLKU7Zn3Gwg3",
  "display": "Patient, Sample"
},
"encounter": {
  "reference": "Encounter/eYa42DgbAQ0XcDw5Rvemy4Er4GT.hL9JoPKWeRZdrPmk3",
  "identifier": {
    "use": "usual",
    "system": "urn:oid:1.2.840.114350.1.13.5826.1.7.3.698084.8",
    "value": "10000602781"
  },
  "display": "Office Visit"
},
"supportingInformation": [
  {
    "reference": "#Indication1"
  },
  {
    "reference": "#OrderSpecificQuestions"
  }
],
"requester": {
  "reference": "Practitioner/eD2q2FEfMDwOc9RRUZDBdEQ3",
  "type": "Practitioner",
  "display": "Starter Provider, MD"
},
"recorder": {
  "reference": "Practitioner/eD2q2FEfMDwOc9RRUZDBdEQ3",
  "type": "Practitioner",
  "display": "Starter Provider, MD"
},
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "699314009",
        "display": "Migraine with persistent visual aura (disorder)"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "346.50",
        "display": "Migraine with persistent visual aura"
      },
      {
```

```
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "G43.509",
        "display": "Migraine with persistent visual aura"
      }
    ],
    "text": "Migraine with persistent visual aura"
  }
],
"reasonReference": [
  {
    "reference":
"Condition/eKpqBsvFtfTDyAxLM1XhK5lMjAeuySrI5aKTwfDX36lRIgomslle8QCIppXrPiS3CAX9tyKvBhTv.1n2cdWaKnA3",
    "type": "Condition",
    "display": "Migraine with persistent visual aura"
  }
],
"dosageInstruction": [
  {
    "timing": {
      "repeat": {
        "boundsPeriod": {
          "start": "2024-03-18T15:37:26Z"
        },
        "frequency": 1,
        "period": 6,
        "periodUnit": "h"
      },
      "code": {
        "text": "Q6H PRN"
      }
    },
    "asNeededBoolean": true,
    "route": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "260548002",
          "display": "Oral (qualifier value)"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.5826.1.7.4.798268.7025",
          "code": "15",
          "display": "Oral"
        }
      ],
      "text": "Oral"
    },
    "doseAndRate": [
      {
        "type": {
          "coding": [
            {
              "system": "http://epic.com/CodeSystem/dose-rate-type",
              "code": "calculated",
              "display": "calculated"
            }
          ],
          "text": "calculated"
        },
        "doseQuantity": {
          "value": 400,
          "unit": "mg",
          "system": "http://unitsofmeasure.org",
          "code": "mg"
        }
      },
      {
        "type": {
          "coding": [
            {
              "system": "http://epic.com/CodeSystem/dose-rate-type",
              "code": "admin-amount",
              "display": "admin-amount"
            }
          ],
          "text": "admin-amount"
        },
        "doseQuantity": {
          "value": 1,
          "unit": "tablet",
          "system": "http://unitsofmeasure.org",
          "code": "{tbl}"
        }
      },
      {
        "type": {
          "coding": [
            {
              "system": "http://epic.com/CodeSystem/dose-rate-type",
```

```
                "code": "ordered",
                "display": "ordered"
              }
            ],
            "text": "ordered"
          },
          "doseQuantity": {
            "value": 400,
            "unit": "mg",
            "system": "http://unitsofmeasure.org",
            "code": "mg"
          }
        }
      ]
    }
  ]
},
```

7.1.6 CDS Hooks ServiceRequest.Create (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/1062-CDS-Hooks-ServiceRequest-Create-Unsigned-Order-R4.html
Epic API ID | 1062
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1062

Description

This web service adds unsigned non-medication orders for a patient, such as procedures or imaging orders. This web service is currently designed for use only with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Sample response

```
{
  "cards": [
    {
      "summary": "CDS Hooks example card summary",
      "selectionBehavior": "at-most-one",
      "suggestions": [
        {
          "label": "Add these actions",
          "actions": [
            {
              "type": "create",
              "description": "Diabetes Order Set",
              "resource": {
                "resourceType": "ServiceRequest",
                "status": "draft",
                "intent": "proposal",
    "extension": [
    {
      "url": "http://fhir.org/argonaut/Extension/pama-rating",
      "valueCodeableConcept": {
        "coding": [
              {
              "system": "http://fhir.org/argonaut/CodeSystem/pama-rating",
              "code": "appropriate"
              }
        ]
      }
    },
    {
      "url": "http://fhir.org/argonaut/Extension/pama-rating-consult-id",
      "valueUri": "urn:uuid:neworder"
    },
    {
      "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-detailed-score",
      "valueCodeableConcept": {
        "coding": [
              {
              "system": "http://open.epic.com/FHIR/StructureDefinition/CodeSystem/pama-rating-detailed-score",
              "code": "11",
              }
        ]
      }
```

```
},
{
  "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-comment",
  "valueString": "Comment from new order"
},
{
  "valueCodeableConcept": {
    "coding": [
       {
         "system": "http://open.epic.com/FHIR/StructureDefinition/CodeSystem/pama-rating-exception",
         "code": ""
       }
    ]
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-exception"
 },
],                          "category": [
              {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/medicationrequest-category",
                    "code": "inpatient",
                    "display": "Inpatient"
                  }
                ]
              }
            ],
            "code": {
              "coding": [
                {
                  "system": "urn:com.epic.cdshooks.action.code.system.orderset-item",
                  "code": "DIABETES"
                }
              ],
              "text": "Diabetes"
            },
"reasonReference": [
  {
    "reference": "Condition/ejwkKH6vIAe5l8Tcdlz1HJbBhhHptXy0VPfxVihFyfMIYuZyi5MBLePwAI.QLCSVuQMgm-
SZww1QNfUs.S5t5xQ3",
    "type": "Condition",
    "display": "Cough"
  },
  {
    "reference": "Condition/ejwkKH6vIAe5l8Tcdlz1HJbBhhHptXy0VPfxVihFyfMJTnDUPRmH6rFLa9S0PUX00KNXN-
YhAmGzPS4Pq.3u-ig3",
    "type": "Condition",
    "display": "Stomach pain"
  }
],
"contained": [
  {
  "resourceType": "Observation",
  "id": "Indication1",
  "status": "preliminary",
  "code": {
    "text": "Indications Of Use"
  },
  "valueString": "headache"
  },
  {
  "resourceType": "Observation",
  "id": "Indication2",
  "status": "preliminary",
  "code": {
    "text": "Indications Of Use"
  },
  "valueString": "sneeze"
  },
  {
  "resourceType": "Observation",
  "id": "IndicationsComment",
  "status": "preliminary",
  "code": {
    "text": "Indications Of Use Comment"
  },
  "valueString": "Comment"
  },
  {
  "resourceType": "QuestionnaireResponse",
  "id": "OrderSpecificQuestions",
  "status": "in-progress",
  "item": [
    {
    "linkId": "Sample Question 1",
    "answer": [
      {
      "extension": [
        {
```

```
            "valueString": "Sample Answer 1",
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-question-comment"
            }
          ],
          "valueString": "Yes"
          }
        ]
        },
      ]
      }
    ]
    }
  ]                "subject": {
                  "reference": "Patient/eorRFcFtkMn3fTS05W1ZlvA3",
                  "display": "Argonaut, Jason"
                }
              }
            }
          ]
        }
      ]
    }
  ]
}
```

7.1.7 CDS Hooks ServiceRequest.Delete (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/10162-CDS-Hooks-ServiceRequest-Delete-Unsigned-Order-R4.html
Epic API ID | 10162
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10162

Description

Sample request

```
{
  "hookInstance": "7f152b27-b80a-11ef-977a-005056be447d",
  "fhirServer": "https://hostname/instance/api/FHIR/R4",
  "hook": "order-select",
  "fhirAuthorization": {
    "access_token": "<opaque access token>",
    "token_type": "Bearer",
    "expires_in": 3600,
    "scope": "<application scopes>",
    "subject": "<your client id>"
  },
  "context": {
    "patientId": "el0VeGTE1vcmdn5N.BKPUFg3",
    "encounterId": "ejHSgHTi-oTHAIeAOzTmnCA3",
    "userId": "PractitionerRole/eg55ZEeKE6a5xlFwDbwiNtA3",
    "selections": [
      "ServiceRequest/eop3o7jtcOP7pPKDMlrwa7v3gIageTU7zf4CY-
    1eSUifnKPg6ZrezJBph5x72Gc6BETZnZSDh.jXoYmz59kZ.yTkwEEiBzKemZrLScH7UYAiKTkV6nBJRsH.R3MvGiKe.UjDMJem2YAEKzQXyd6Eztb1hY
    2Key0aUpAWktSReTy8NTxA09UUUr273Sh69gYtb3"
    ],
    "draftOrders": {
      "resourceType": "Bundle",
      "type": "collection",
      "entry": [
        {
          "resource": {
            "resourceType": "ServiceRequest",
            "id": "eop3o7jtcOP7pPKDMlrwa7v3gIageTU7zf4CY-
    1eSUifnKPg6ZrezJBph5x72Gc6BETZnZSDh.jXoYmz59kZ.yTkwEEiBzKemZrLScH7UYAiKTkV6nBJRsH.R3MvGiKe.UjDMJem2YAEKzQXyd6Eztb1hY
    2Key0aUpAWktSReTy8NTxA09UUUr273Sh69gYtb3",
            "contained": [
              {
                "resourceType": "QuestionnaireResponse",
                "id": "OrderSpecificQuestions",
                "status": "in-progress",
                "item": [
                  {
                    "linkId": "RIS SEDATION NEED OSQ"
                  }
                ]
```

```
            }
          ],
          "status": "draft",
          "intent": "order",
          "category": [
            {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "416800000",
                  "display": "Inpatient"
                }
              ],
              "text": "Inpatient"
            }
          ],
          "code": {
            "coding": [
              {
                "system": "http://www.ama-assn.org/go/cpt",
                "code": "70470",
                "display": "CHG CT SCAN HEAD COMBO"
              }
            ],
            "text": "CT Head w and wo IV Contrast"
          },
          "quantityQuantity": {
            "value": 1
          },
          "subject": {
            "reference": "Patient/el0VeGTE1vcmdn5N.BKPUFg3",
            "display": "Willow, Eric E"
          },
          "encounter": {
            "reference": "Encounter/ejHSgHTi-oTHAIeAOzTmnCA3",
            "identifier": {
              "use": "usual",
              "system": "urn:oid:1.2.840.114350.1.13.0.1.7.3.698084.8",
              "value": "15249"
            },
            "display": "Hospital Encounter"
          },
          "occurrenceTiming": {
            "event": [
              "2024-12-11T21:55:00Z"
            ],
            "repeat": {
              "boundsPeriod": {
                "start": "2024-12-11T21:55:00Z",
                "end": "2024-12-11T21:55:00Z"
              },
              "count": 1
            },
            "code": {
              "text": "Once"
            }
          },
          "requester": {
            "reference": "Practitioner/eM5CWtq15N0WJeuCet5bJlQ3",
            "type": "Practitioner",
            "display": "Physician Family Medicine, MD"
          },
          "supportingInfo": [
            {
              "reference": "#OrderSpecificQuestions"
            }
          ]
        }
      }
    ]
  }
 },
 "extension": {
   "com.epic.cdshooks.request.bpa-trigger-action": "18",
   "com.epic.cdshooks.request.cds-hooks-specification-version": "1.1",
   "com.epic.cdshooks.request.fhir-version": "R4",
   "com.epic.cdshooks.request.criteria-id": "504",
   "com.epic.cdshooks.request.epic-version": "11.2",
   "com.epic.cdshooks.request.cds-hooks-implementation-version": "1.8",
   "com.epic.cdshooks.request.cds-hooks-su-version": "0",
   "internal.epic.cdshooks.request.epic-emp-id": "10000"
 }
}
```

Sample response

```
{
  "cards": [
    {
```

```
      "uuid": "96918b3c-c068-4f4d-a2af-a573f6194519",
      "summary": "Lifetime Radiation Exposure Exceeded",
      "detail": "This patient has exceeded their lifetime dose of radiation exposure.",
      "indicator": "warning",
      "extension": {
        "com.epic.cdshooks.card.detail.content-type": "text/html"
      },
      "source": {
        "label": "Click here for ACR Radiations Guidelines.",
        "url": "http://www.ncbi.nlm.nih.gov/pubmed/21716271",
    "topic": {
      "code": "RadiationExposureExceeded"
    }
      },
      "suggestions": [
        {
          "uuid": "a2b24553-ae4b-4885-b067-b1e98a2fecc0",
          "label": "Remove CT Head w and wo IV Contrastx",
          "isRecommended": true,
          "actions": [
            {
              "type": "delete",
              "description": " ",
              <b>"resourceId": "ServiceRequest/eop3o7jtcOP7pPKDMlrwa7v3gIageTU7zf4CY-
    1eSUifnKPg6ZrezJBph5x72Gc6BETZnZSDh.jXoYmz59kZ.yTkwEEiBzKemZrLScH7UYAiKTkV6nBJRsH.R3MvGiKe.UjDMJem2YAEKzQXyd6Eztb1hY
    2Key0aUpAWktSReTy8NTxA09UUUr273Sh69gYtb3"</b>
            }
          ]
        }
      ],
      "selectionBehavior": "any"
    }
  ]
}
```


7.1.8 CDS Hooks ServiceRequest.Read (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/10032-CDS-Hooks-ServiceRequest-Read-Unsigned-Order-R4.html
Epic API ID | 10032
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/ServiceRequest/{ID}
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10032

Description

This web service reads unsigned procedure order information from a patient's chart. This web service is currently designed for use only with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR ServiceRequest ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
ServiceRequest | Resource | False | | | An instance of the R4 ServiceRequest resource.
ServiceRequest.asNeededBoolean | Boolean | True | | | If set to true, indicates that the order is only performed when needed within the specified schedule.
ServiceRequest.basedOn | Reference | True | Yes | | Identifier for the procedure if it is part of a panel.
ServiceRequest.category | CodeableConcept | True | Yes | | The category of the procedure request. Supported categories: inpatient outpatient
ServiceRequest.code | CodeableConcept | True | | | External codings for procedures linked to the order. The API returns all possible codings (SNOMED, CPT, LOINC, CBV, THL, Kuntalitto) associated with the Procedure/Service in Epic.
ServiceRequest.contained | ResourceContainer | True | Yes | | Supported resources: Observation QuestionnaireResponse
ServiceRequest.encounter | Reference | True | | | Reference to the encounter resource in which the ServiceRequest was placed.
ServiceRequest.id | id | False | | | FHIR ServiceRequest ID.
ServiceRequest.identifier | Identifier | True | Yes | | Identifier for the unsigned order.
ServiceRequest.intent | code | False | | | The intent code. Epic supported values include: order
ServiceRequest.extension (also: is-scannable-encounter) | Extension | True | Yes | | If true, this order or procedure is scannable.

ServiceRequest.note | Annotation | True | Yes | | Comments on the ServiceRequest.
ServiceRequest.occurrenceTiming | Timing | True | | | This event element is populated for auto-released orders.
ServiceRequest.extension (also: order-priority-category) | Extension | True | Yes | | The priority of the procedure.
ServiceRequest.extension (also: pama-rating) | Extension | True | Yes | | The appropriateness rating of the procedure.
ServiceRequest.extension (also: pama-rating-comment) | Extension | True | Yes | | The comment of the procedure relating to its appropriateness.
ServiceRequest.extension (also: pama-rating-consult-ID) | Extension | True | Yes | | The session ID of the procedure.
ServiceRequest.extension (also: pama-rating-detailed-score) | Extension | True | Yes | | The numeric score of the procedure.
ServiceRequest.extension (also: pama-rating-exception) | Extension | True | Yes | | The exception of the procedure relating to its appropriateness.
ServiceRequest.extension (also: pama-rating-instant-utc) | Extension | True | Yes | | The instant the procedure's decision support was shown.
ServiceRequest.performer | Reference | True | Yes | | The requested performer for the procedure.
ServiceRequest.priority | String | True | | | The priority. The value returned here depends on custom mapping configuration completed by the Epic organization.
ServiceRequest.reasonCode | CodeableConcept | True | Yes | | The standard terminology codes mapped to the procedure, which can include LOINC, SNOMED, CPT, CBV, THL, or Kuntalitto codes.
ServiceRequest.reasonReference | Reference | True | Yes | | Reference to the Condition resource.
ServiceRequest.requester | Reference | True | | | Who or what is requesting the service. Reference to a Practitioner resource.
ServiceRequest.status | code | False | | | The status of the request. Supported values include: draft
ServiceRequest.subject | Reference | False | | | Reference to a patient resource for whom the ServiceRequest was made.
ServiceRequest.supportingInfo | Reference | True | Yes | | Additional clinical information. Reference to contained resources.

Sample request

```
Sample not available
```

Sample response


```
{
  "resource": {
    "resourceType": "ServiceRequest",
    "id": "eol4pYmBbVEjKkMoKbhBR75dWU8HNtWVOmdMv4p8DrOyWrhwQLISA2MvZoI-INiw6jqrtpAZfPFenfM0S.2DkLZh5WTfqSsuaKgcIA65T40P0
    d4yPPi5Ec94hymlHbFHQBlIl9FBx-g1Gi68I.CNpPt1vVNawGtLThiTZX8r4d9K5xnBJIdgFIjPk-VJzI9vfWYXJJ5ONmJtFJrlhUwM3uA3",
    "contained": [
      {
        "resourceType": "Observation",
        "id": "Indication1",
        "status": "preliminary",
        "code": {
          "text": "Indications Of Use"
        },
        "valueString": "LIMB INJURY^10157"
      },
      {
        "resourceType": "QuestionnaireResponse",
        "id": "OrderSpecificQuestions",
        "status": "in-progress",
        "item": [
          {
            "linkId": "DEGREE OF PAIN",
            "answer": [
              {
                "valueString": "MODERATE"
              }
            ]
          }
        ]
      }
    ],
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://fhir.org/argonaut/CodeSystem/pama-rating",
              "code": "appropriate"
            }
          ]
        },
        "url": "http://fhir.org/argonaut/Extension/pama-rating"
      },
      {
        "valueUri": "dss-session at timestamp 3/19/2024 9:20:01 AM",
        "url": "http://fhir.org/argonaut/Extension/pama-rating-consult-id"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/CodeSystem/pama-rating-detailed-score",
              "code": "11",
              "display": "Appropriate"
            }
          ]
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-detailed-score"
```

```
  },
  {
    "valueString": "dss-comment at timestamp 3/19/2024 9:20:01 AM",
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-comment"
  },
  {
    "valueInstant": "5781766850",
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-instant-utc"
  },
  {
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.5826.1.7.10.798268.120",
          "code": "6",
          "display": "Routine"
        }
      ]
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-priority-category"
  }
],
"identifier": [
  {
    "type": {
      "text": "MPI ID"
    },
    "system": "http://open.epic.com/FHIR/StructureDefinition/cds-hooks-unsigned-order-mpi-id/399"
  }
],
"basedOn": [
  {
    "identifier": {
      "system": "http://open.epic.com/FHIR/StructureDefinition/cds-hooks-unsigned-order-panel-id",
      "value": "O89953"
    }
  }
],
"status": "draft",
"intent": "order",
"category": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "416800000",
        "display": "Inpatient"
      }
    ],
    "text": "Inpatient"
  }
],
"priority": "routine",
"code": {
  "coding": [
    {
      "system": "http://www.ama-assn.org/go/cpt",
      "code": "73060",
      "display": "X-RAY HUMERUS"
    }
  ],
  "text": "X-RAY HUMERUS"
},
"quantityQuantity": {
  "value": 1
},
"subject": {
  "reference": "Patient/emFja7sdXmfbww45i6PKxPQ3",
  "display": "Parikh, Dwight"
},
"encounter": {
  "reference": "Encounter/eRSxWe5QCIuYnXAO7m2z3p5LKfY3Z6A65bmj4LvfT9LE3",
  "identifier": {
    "use": "usual",
    "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
    "value": "10003005345"
  },
  "display": "Office Visit"
},
"occurrenceTiming": {
  "event": [
    "2024-03-19T14:19:00Z"
  ],
  "repeat": {
    "boundsPeriod": {
      "start": "2024-03-19T14:19:00Z",
      "end": "2024-03-19T14:19:00Z"
    },
    "count": 1
```

```
      },
      "code": {
        "text": "One Time"
      }
    },
    "requester": {
      "reference": "Practitioner/eD2q2FEfMDwOc9RRUZDBdEQ3",
      "type": "Practitioner",
      "display": "Vishal Parikh, MD"
    },
    "supportingInfo": [
      {
        "reference": "#Indication1"
      },
      {
        "reference": "#OrderSpecificQuestions"
      }
    ]
  }
}
```

Change log

Epic Version	Description	Type
August 2024 (with SU to February 2024)	Procedure MPI ID, if configured, will now be returned in the code element. Previously, this value was returned in the identifier element.	Breaking change

7.1.9 CDS Hooks ServiceRequest.Update (Unsigned Order) (R4)

File: site/apis/07-CDS-Hooks/1-R4/10643-CDS-Hooks-ServiceRequest-Update-Unsigned-Order-R4.html
Epic API ID | 10643
FHIR version | R4
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10643

Description

Epic's CDS Hooks specification

Sample response

```
"systemActions": [
{
    "type": "update",
  "resource": {
      "resourceType": "ciBpaEchoProperty:$.context.draftOrders.entry[0].resource.resourceType",
            "id": "ciBpaEchoProperty:$.context.draftOrders.entry[0].resource.id",
    "extension": [
     {
        "valueCodeableConcept": {
        "coding": [
           {
             "system": "http://open.epic.com/FHIR/StructureDefinition/CodeSystem/pama-rating-exception",
             "code": ""
           }
        ]
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-exception"
     },
      {
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-comment",
        "valueString": ""
      },
      {
            "url": "http://fhir.org/argonaut/Extension/pama-rating-consult-id",
            "valueUri": ""
      },
      {
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/pama-rating-detailed-score",
            "valueCodeableConcept": {
            "coding": [
                  {
                    "system": "http://open.epic.com/FHIR/StructureDefinition/CodeSystem/pama-rating-detailed-score",
                    "code": "",
                  }
            ]
            }
      },
      {
```

```
            "url": "http://fhir.org/argonaut/Extension/pama-rating",
            "valueCodeableConcept": {
            "coding": [
                  {
                    "system": "http://fhir.org/argonaut/CodeSystem/pama-rating",
                    "code": ""
                  }
            ]
            }
      }
    ],
            "contained": [
              {
                "resourceType": "Observation",
                "id": "Indication1",
                "status": "preliminary",
                "code": {
                  "text": "Indications Of Use"
                },
                "valueString": "JRA"
              },
      {
                "resourceType": "Observation",
                "id": "IndicationsComment",
                "status": "preliminary",
                "code": {
                  "text": "Indications Of Use Comment"
                },
                "valueString": "This is a comment."
              },
              {
                "resourceType": "QuestionnaireResponse",
                "id": "OrderSpecificQuestions",
                "status": "in-progress",
                "item": [
                  {
                    "linkId": "free-text",
        "answer": [
          {
            "extension": [
            {
              "valueString": "This is a comment",
              "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-question-comment"
            }
            ],
            "valueString": "This is free text",
          },
          {
            "extension": [
            {
              "valueString": "This is the second comment",
              "url": "http://open.epic.com/FHIR/StructureDefinition/extension/order-question-comment"
            }
            ],
            "valueString": "This is the second answer",
          }
        ],
                  },
                ]
              }
            ],
            "status": "draft",
            "intent": "order",
            "priority": "routine",
            "supportingInfo": [
              {
                "reference": "#OrderSpecificQuestions"
              }
            ]
  }
}
]
```

7.2 CDS Hooks: STU3

File: site/apis/07-CDS-Hooks/2-STU3/index.html

* CDS Hooks Condition.Create (Encounter Diagnosis, Problems) (STU3)
* CDS Hooks MedicationRequest.Create (Unsigned Order) (STU3)
* CDS Hooks MedicationRequest.Delete (Unsigned Order) (STU3)
* CDS Hooks ProcedureRequest.Create (Unsigned Order) (STU3)
* CDS Hooks ProcedureRequest.Delete (Unsigned Order) (STU3)
* CDS Hooks ProcedureRequest.Update (Unsigned Order) (STU3)

7.2.1 CDS Hooks Condition.Create (Encounter Diagnosis, Problems) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/1059-CDS-Hooks-Condition-Create-Encounter-Diagnosis-Problems-STU3.html
Epic API ID | 1059
FHIR version | STU3
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
Keywords | Health Concerns;Problems;Problem List
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1059

Description

This web service adds problems to a patient's chart. This web service is currently designed only for use with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

This web service works similar to the Condition.Create (STU3) web service except it does not currently support the following fields: onsetDateTime, onSetPeriod, clinicalStatus, text.

7.2.2 CDS Hooks MedicationRequest.Create (Unsigned Order) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/1061-CDS-Hooks-MedicationRequest-Create-Unsigned-Order-STU3.html
Epic API ID | 1061
FHIR version | STU3
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1061

Description

This web service adds unsigned medication orders for a patient. This web service is currently designed only for use with CDS Hooks integrations and should not be used for other purposes. When used in a CDS Hooks context, this web service suggest follow-up medications orders in the CDS card displayed to the user. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

7.2.3 CDS Hooks MedicationRequest.Delete (Unsigned Order) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/10161-CDS-Hooks-MedicationRequest-Delete-Unsigned-Order-STU3.html
Epic API ID | 10161
FHIR version | STU3
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10161

Description

7.2.4 CDS Hooks ProcedureRequest.Create (Unsigned Order) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/1063-CDS-Hooks-ProcedureRequest-Create-Unsigned-Order-STU3.html
Epic API ID | 1063
FHIR version | STU3
Categories | CDS Hooks

API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1063

Description

This web service adds unsigned non-medication orders for a patient, such as procedures or imaging orders. This web service is currently designed for use only with CDS Hooks integrations and should not be used for other purposes. See Epic's CDS Hooks specification for more information on creating an integration that utilizes CDS Hooks.

Sample response

```
{
    "cards": [{
            "summary": "CDS Hooks Card",
            "detail": "Details for CDS Hooks Card",
            "uuid": "UUID for Card",
            "source": {
                "label": "Sourced from Epic's Vendor Services Unit Test endpoint",
                "icon": "https://www.epic.com/favicon.ico",
                "url": "https://appmarket.epic.com/api/TestCDSHooks"
            },
            "links": [{
                    "label": "SMART App",
                    "url": "https://appmarket.epic.com/Test/smart",
                    "type": "smart",
                    "appContext": "b516b58a-dbde-4888-a8a2-24173d63adcb" //Context to tie CDS instance to SMART on FHIR
    app launch
                }
            ],
            "suggestions": [{
                    "label": "Create Procedure Order", //Unused
                    "uuid": "create-proc-order", //Should be implemented as guid
                    "isRecommended": false, //Determines if automatically selected as follow-up order or if user has to
    manually select to order
                    "actions": [{
                            "type": "create", //always "create"
                            "description": "Create Procedure Order", //Used as display next to "Order"/"Do Not Order"
    next to description of order from mapped preference list item
                            "resource": {
                                "code": {
                                    "text": "SpO2 Monitoring", //Unused
                                    "coding": [{
                                            "code": "1", //Mapped to an "Import Key" on preference list in CDS
    Configuration
                                            "system": "urn:com.epic.cdshooks.action.code.system.preference-list-item"
                                        }
                                    ]
                                },
                                "intent": "proposal", //always "proposal"
                                "status": "draft", //always "draft"
                                "category": [{
                                        "coding": [{
                                                "code": "inpatient", //can be "inpatient" or "outpatient"
                                                "system": "http://terminology.hl7.org/CodeSystem/medicationrequest-
    category", //This uses medicationrequest-category to utilize an existing FHIR mapping for inpatient/outpatient
                                                "display": "Inpatient"
                                            }
                                        ]
                                    }
                                ],
                                "resourceType": "ProcedureRequest", //Required
                                "subject": {
                                    "reference": "Patient/el0VeGTE1vcmdn5N.BKPUFg3" //Patient FHIR ID required
                                }
                            }
                        }
                    ]
                }
            ]
        }
    ]
}
```

7.2.5 CDS Hooks ProcedureRequest.Delete (Unsigned Order) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/10163-CDS-Hooks-ProcedureRequest-Delete-Unsigned-Order-STU3.html
Epic API ID | 10163
FHIR version | STU3

Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10163

Description

7.2.6 CDS Hooks ProcedureRequest.Update (Unsigned Order) (STU3)

File: site/apis/07-CDS-Hooks/2-STU3/10644-CDS-Hooks-ProcedureRequest-Update-Unsigned-Order-STU3.html
Epic API ID | 10644
FHIR version | STU3
Categories | CDS Hooks
API type | Industry-Standard
USCDI | No
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10644

Description

Epic's CDS Hooks specification